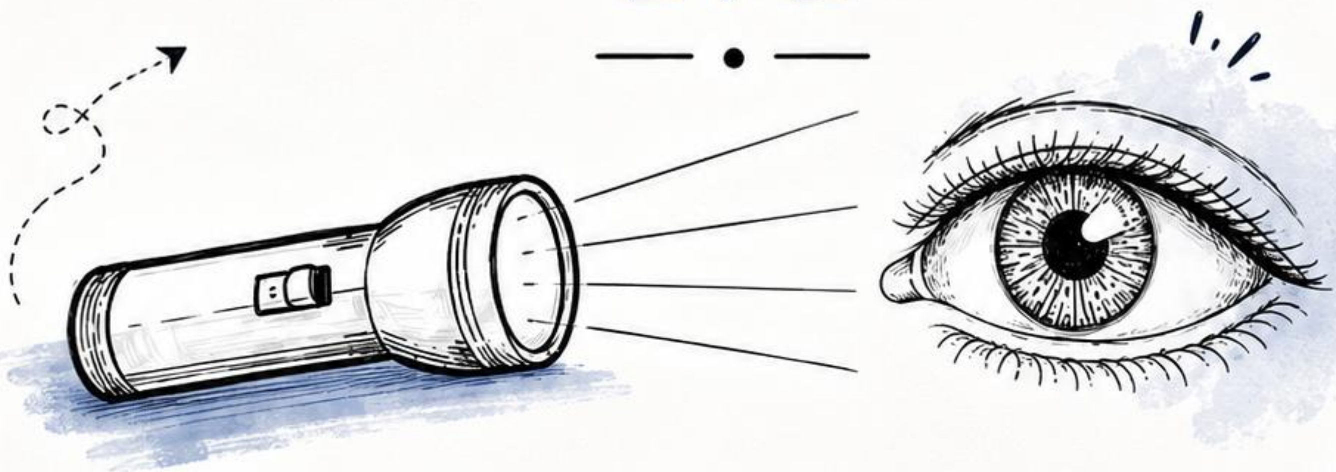
PHYSICS

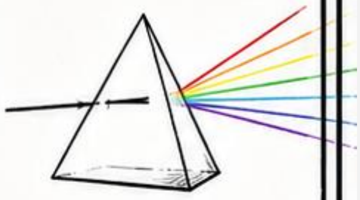
LIGHT

EYE



Class 10

-by ERSHAD SIR



Ray Optics &Optical Instrument

Q1. The human eye (retina) with the sensitivity to detect E.M. waves within a --

A. Large range of the E.M. spectrum
B. Small range of the E.M. spectrum
C. Both (A) & (B)
D. None of these

Ans. B

Q2. Electromagnetic radiation belonging to the sensitivity of the retina of the E.M. spectrum is called—

A. X- Rays
B. U.V. Rays
C. Light
D. Radio waves

Ans. C

Q3. E.M. radiation belonging to light is of wavelength of about—

A. 400 nm to 750 nm
B. 450 nm to 750 nm
C. 800 nm to 1000 nm
D. 500 nm to 750 nm

Ans. A

Q4. Define Light?

ANS. The human eye (retina) with sensitivity to detect E.M. waves within a small range of the E.M. spectrum. E.M. radiation belonging to this region of the spectrum (wavelength of about 400 nm to 800 nm) is called Light.

Q5. Visible Part of the E.M. Spectrum

A. X-Rays
B. Radio Waves
C. U.V. Rays
D. Light

Ans. D

→ Light is an electromagnetic wave of wavelength belonging to the visible part of the spectrum.

→ The wavelength of light is very small compared to the size of ordinary objects that we encounter commonly. (Generally, of the order of a few E.M. or larger)

Q6. Define Ray of Light?

ANS. A light wave can be considered to travel from one point to another along a straight line joining them. The path is called a ray of light.

Q7. Define Beam of Light

ANS. A bundle of rays of light constitutes a beam of light.

→ The particle model of light is also known as the corpuscular model of light or corpuscular theory of light.

→ Corpuscles, which are now understood as photons.

→ Newton developed the corpuscular model of light proposed by Descartes.

Q8. Define Corpuscles/Photons

ANS. The particle model of light presumes that light energy is concentrated in tiny particles called corpuscles (photons).

→ Photons (corpuscles) of light were massless elastic particles.

→ Newton's understanding of mechanics led him to formulate a simple model of reflection and refraction.

→ A ball bouncing from a smooth plane surface obeys the laws of reflection.

→ When a ball (particle of light/photons) undergoes an elastic collision, the magnitude of the velocity remains the same. Only the velocity component perpendicular to the surface changes; the momentum gets reversed in reflection.

→ Newton argued that smooth surfaces like mirror Reflects the Corpuscles in a similar manner.

Q9. Which Components of Momentum of Light Particle (Photon) Get Reversed?

A. Only Parallel Component.
B. Only the normal component.
C. Both (A) & (B)
D. None of these **ANS. B**

→ When the particle of light is in elastic collision, the magnitude of the velocity remains the same.

→ As the surface is smooth, there is no force acting parallel to the surface, so the component of momentum in this direction also remains the same.

Q10. Define Angle of Reflection

ANS. The angle between the reflected ray and the normal to the reflecting surface or mirror is called the angle of reflection.

Q11. Define Angle of Incidence

ANS. The angle between the incident ray and the normal to the reflecting surface or mirror is called the angle of incidence.

LAW OF REFLECTION

1) ***The angle of incidence is equal to the angle of reflection, i.e.,***

$$\angle i = \angle r$$

2) ***The incident ray, reflected ray, and the normal to the reflecting surface at the point of incidence lie in the same plane.***

→ The laws of reflection are valid at each point on any reflecting surface, whether plane or curved.

Q12. Define Normal of the Spherical Surface

ANS. The normal on the surface is taken as the normal to the tangent to the surface at the point of incidence. That is, the normal is along the radius, the line joining the centre of curvature of the mirror to the point of incidence.

Q13. Define Pole

ANS. The geometric centre of a spherical mirror is called the pole of the mirror. It is denoted by P.

Q14. Define Optical Centre

ANS. The geometric centre of a spherical lens is called the optical centre. It is denoted by O.

Q15. Define Principal Axis of a Spherical Mirror

ANS. The straight line passing through the pole and the centre of curvature of a spherical mirror is called its principal axis.

Q16. Define Principal Axis of a Spherical Lens

ANS. The straight line passing through the optical centre and the centres of curvature of both spherical surfaces of a lens is called the principal axis.

Q17. Define Paraxial Rays

ANS. The rays of light which are very close to and nearly parallel to the principal axis are called paraxial rays.

Q18. Define Principal Focus of a Concave Mirror

ANS. The point on the principal axis at which rays parallel to the principal axis actually meet after reflection from a concave mirror is called the principal focus **of the concave mirror.**

Q19. Define Principal Focus of a Convex Mirror

ANS. The point on the principal axis from which rays parallel to the principal axis appear to diverge after reflection from a convex mirror is called the principal focus of the convex mirror.

Q20. Define Focal Plane of a Mirror

ANS. If a parallel paraxial beam of light is incident making some angle with the principal axis, the reflected rays converge at a point in a plane through F normal to the principal axis. This plane is called the focal plane of the mirror.

Q21. Define Focal Length of the Mirror

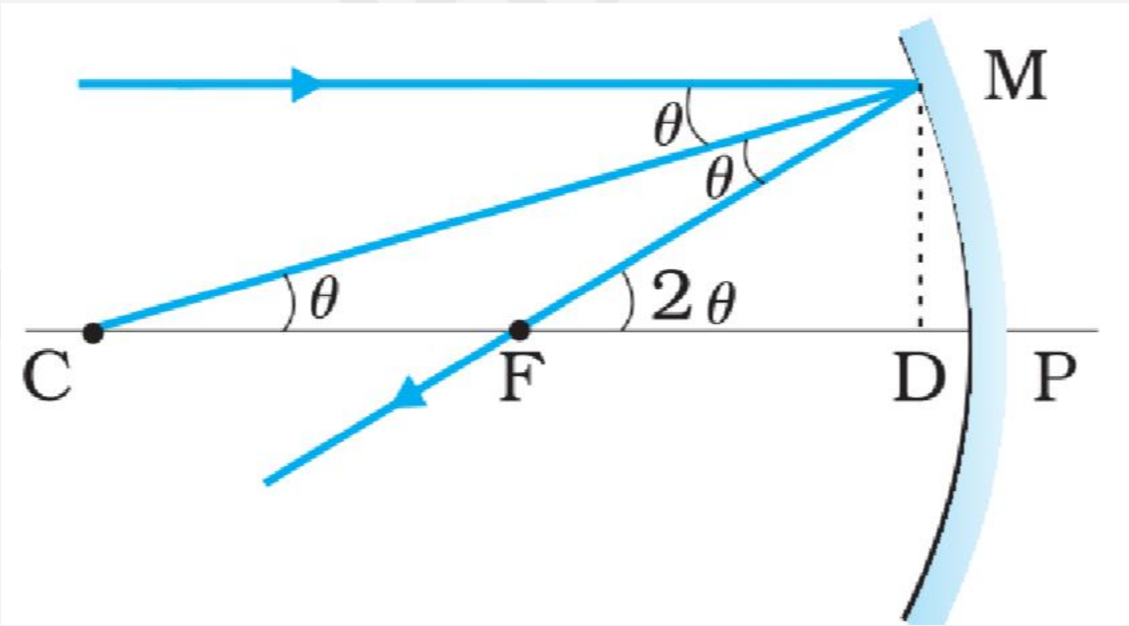
ANS. The distance between the focus F and the pole P of the mirror is called the focal length of the mirror.

- The focal length of the mirror is denoted by f.
- $f = \frac{R}{2}$, where R is the radius of curvature of the mirror.

or

- $R = 2f$

Q22. Derive the Relation Between f and R



- Given, ∠MCP = θ & ∠MFP = 2θ
-
- $\tan\theta = \frac{MD}{CD}$ & $\tan 2\theta = \frac{MD}{FD}$
- For small θ (paraxial rays),

- $\tan\theta \approx \theta$ & $\tan 2\theta \approx 2\theta$
- Therefore,
- $2\theta = \frac{MD}{FD}$

$$\theta = \frac{MD}{CD}$$

Hence,

- $\frac{2MD}{CD} = \frac{MD}{FD}$

$$\frac{2}{CD} = \frac{1}{FD}$$

For small θ, point D is very close to point P.

- Therefore,
- $FD = f$
- and
- $CD = R$
- Hence,
- $f = \frac{R}{2}$ or $R = 2f$

Q23. Define Image

ANS. If rays emanating from a point actually meet at another point after reflection and/or refraction, that point is called the image of the first point.

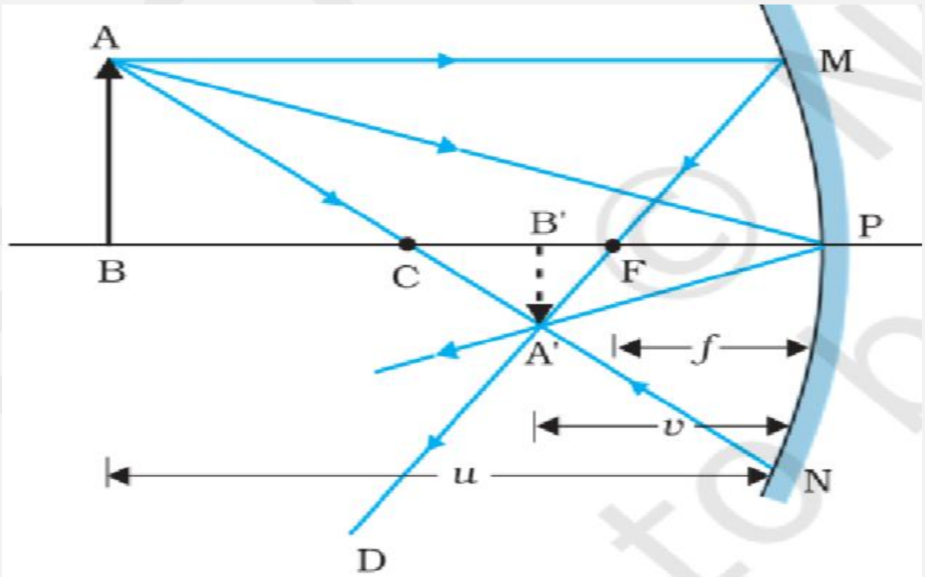
- ➔ The image is **real** if the rays actually converge to the point.
- ➔ The image is **virtual** if the rays do not actually meet but appear to diverge from the point when produced backward.
- ➔ An image is thus a point-to-point correspondence with the object established through reflection and/or refraction.
- ➔ The mirror equation or mirror formula gives the relation between the object distance **(u)**, image distance **(v)** and focal length **(f)**.

Q24. Define Mirror Formula

ANS. **The sum of the reciprocals of the object distance and image distance is equal to the reciprocal of the focal length.**

$$\boxed{\frac{1}{f} = \frac{1}{u} + \frac{1}{v}}$$

Q25. Obtain Mirror Formula



- In right-angled triangles,
- $\frac{B'A'}{PM} = \frac{B'F}{FP}\ (i)$
  In right-angled triangles A'B'P and ABP,
-
-
- $\frac{B'A'}{BA} = \frac{B'P}{BP}\ (ii)$

- Comparing (i) and (ii),
- $\frac{B'F}{FP} = \frac{B'P}{BP}$

$$\frac{B'P - FP}{FP} = \frac{B'P}{BP}$$

Substituting distances,

- $\frac{v-f}{f} = \frac{v}{u}$

Therefore,

- $\boxed{\frac{1}{f} = \frac{1}{u} + \frac{1}{v}}$

- This relation is known as the mirror equation.

Q26. Define Linear Magnification (m)

ANS. Linear magnification (m) is the ratio of the height of the image (h') to the height of the object (h).

$$m = \frac{h'}{h}$$

Q27. Derive the Formula of Linear Magnification

ANS. In triangles A'B'P and ABP, (see above figure…)

$$\frac{B'A'}{BA} = \frac{B'P}{BP}$$

Therefore,

$$-\frac{h'}{h} = -\frac{v}{u}$$

Hence,

$$\boxed{m = \frac{h'}{h} = -\frac{v}{u}}$$

REFRACTION

- When a beam of light encounters another transparent medium a part of light gets reflected back into the first medium while the rest enters the other (second medium).

Q28. Define Refraction of Light.

ANS. The direction of propagation of an obliquely incident ray of light, that enters the other medium, changes at the interface of the two media. This phenomenon is called refraction.

Snell's Law of Refraction (Experimentally)

3) ***The incident ray, the refracted ray, and the normal to the interface at the point of incidence all lie in the same plane.***
4) ***The ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant.***
5) $\frac{\sin i}{\sin r} = n_{21}$,
6) ***where n_{21} is a constant called the refractive index of the second medium with respect to the first medium.***

- Refractive index is a characteristic of the pair of media.
- Refractive index depends on the wavelength of light.
- Refractive index is independent of the angle of incidence.
- If $n_{21} > 1$, then $r < i$; the refracted ray bends towards the normal. In such a case medium 2 is optically denser than medium 1.
- If $n_{21} < 1$, then $r > i$; the refracted ray bends away from the normal. This happens when light goes from a denser medium to a rarer medium.
- Mass density of an optically denser medium may be less than that of an optically rarer medium.
- For e.g.; Turpentine and Water.
- Mass density of turpentine is less than that of water, but its optical density is higher.

Q29. Define Optical Density.

ANS. Optical density is the ratio of the speed of light in two media.

$$n_{12} = n_{21}$$

- It is stated that $n_{32} = n_{31} \times n_{12}$, where n_{31} represents the refractive index of medium 3 with respect to medium 1.

Some elementary results based on the laws of refraction are:

i) Rectangular glass slab
ii) Apparent depth and real depth
iii) Apparent position of the sun
iv) Advance sunrise and delayed sunset

i) Rectangular Glass Slab

For a rectangular slab, refraction takes place at two interfaces:

a. Air → Glass
b. Glass → Air

Angle of incidence = Angle of emergence

$$i = e$$

Hence, the emergent ray is **parallel** to the incident ray.

There is **no angular deviation**, but the ray undergoes **lateral displacement (lateral shift)** with respect to the incident ray.

ii) Apparent Depth and Real Depth

For viewing near the normal direction, it can be shown that

Apparent depth = Real depth ÷ Refractive index of the medium

or,

$$\text{Real depth} = \mu \times \text{Apparent depth}$$

Hence,

$$\mu = \frac{\text{Real depth}}{\text{Apparent depth}}$$

iii) (iii) & (iv) Refraction through the Atmosphere

The refraction of light through the atmosphere is responsible for many interesting phenomena:

1. *The Sun is visible a little before the actual sunrise.*
2. *The Sun is visible a little after the actual sunset.*

Actual sunrise means the actual crossing of the horizon by the Sun.

The refractive index of air with respect to vacuum is approximately 1.0003.

Due to this, the apparent shift in the direction of the Sun is about half a degree, and the corresponding time difference between the actual and apparent sunrise/sunset is about 2 minutes.

- The apparent flattening (oval shape) of the actual sun at sunset and sunrise is also due to the same phenomenon.
- Whether it is a wave or a particle or a human being, whenever two medium and two velocities are involved, one must follow Snell's law if one wants to take the shortest time, i.e.,
- $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = n$
-
- In the case of light, v_1/v_2, the ratio of the velocity of light in vacuum to that in medium, is the refractive index of the medium n.

Q30. Define internal reflection.

ANS. **When light travels from an optically denser medium to a rarer medium, at the interface, it is partly reflected back into the same medium and partly refracted to the second medium. This is called the internal reflection.**

-
- Angle of refraction is larger than angle of incidence when a ray of light travels from denser to rarer then,
  $n_{21} < 1$
- Angle of refraction is larger than the angle of incidence when the refracted ray is bend away the normal.
- Angle of incidence is larger than angle of refraction when a ray of light travels from rarer to denser then,
  $n_{21} > 1$
- Angle of incidence is larger than angle of refraction when a refracted ray of light bend towards the normal.

In Short

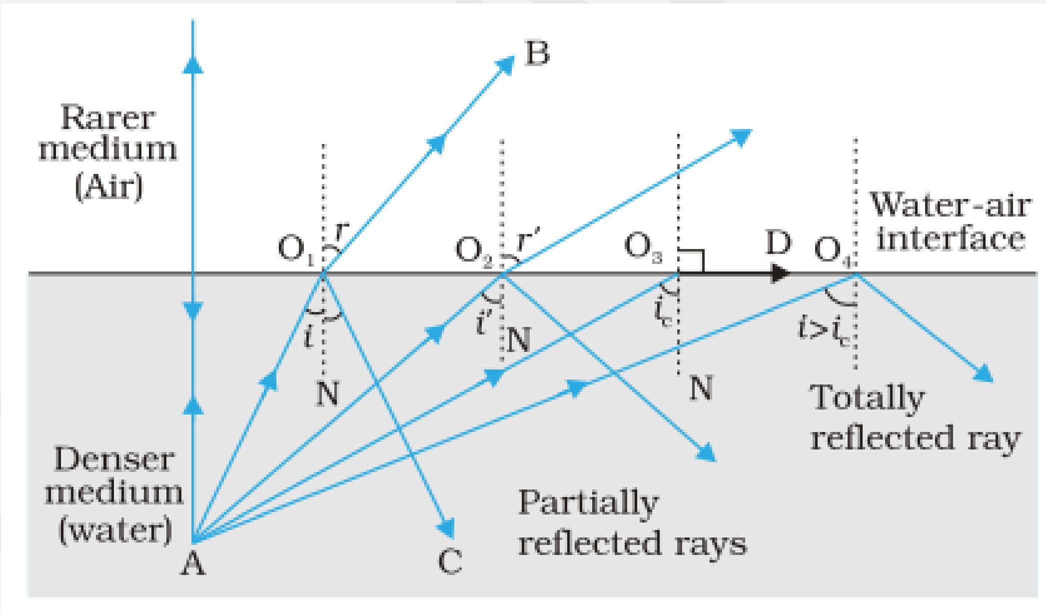
If $n_{21} > 1$, then

I) Ray of light travels from rarer to denser medium.
II) Refracted ray bends toward the normal.
III) $\angle I > \angle r$

if $n_{21} < 1$, then

IV) Ray of light travels from denser to rarer.
V) Refracted ray bends away from the normal.
VI) $\angle I < \angle r$

- Snell's law of refraction is true for angle $0° < I < 90°$
- As the angle of incidence increases, so does the angle of refraction, till for the ray AO_3, the angle of refraction is $\pi/2$. ($\pi = 180°$)
- The refracted ray is bent so much away from the normal that it grazes the surface at the interface b/w the two media. This is shown by the ray AO_3D.



- If the angle of incidence is increased still further (e.g. Ray AO4), refraction is not possible and the incident ray is totally reflected. This is called Total Internal Reflection.
- When light gets reflected by a surface, normally some fraction of it get transmitted.
- The reflected ray therefore, is always less intense than the incident ray, however smooth the reflecting surface may be.
- In total internal reflection, on the other hand, no transmission of light takes place.

Q31. Define critical angle (ic).

ANS. The angle of incidence corresponding to an angle of refraction 90°, say ∠AO4N, is called the critical angle (ic) for the given pair of media.

- $\frac{\sin i}{\sin r} = n_{21}$ if the relative refractive index is less than one then, since maximum value of sin r is unity, there's an upper limit to the value of sin i for which the [unclear] can be satisfied that is
  i = ic such that
  ∴ Sin ic = n_{21}

- For value of i larger than ic, Snell's law of refraction cannot be satisfied and hence no refraction is possible.
- The refractive index of denser medium 1 w.r.t rarer medium 2 will be
  n_{12} = 1/Sin i
- The path of beam inside the soap solution/water shines brightly.

Total internal reflection in nature and its technological applications.

1. **MIRAGE**
   - On hot summer days, the air near the ground becomes hotter than the air at higher levels.
   - The refractive index of air increases with its density.
   - Hotter air is less dense, and has smaller refractive index than the cooler air.

   - If the air currents are small, that is, the air is still, the optical density at different layers of air increases with height.
   - As a result, light from a tall object such as a tree, passes through a medium whose refractive index decreases towards the ground.

   ****** Thus, a ray of light from such an object successively bends away from the normal and undergoes total internal reflection, if the angle of incidence for the air near the ground exceeds the critical angle.***

   -

- ➔ To a distant observer, the light appears to be coming from somewhere below the grounds.
- ➔ The observer naturally assumes that light is being reflected from the ground, say by a pool of water near the tall object, such inverted image of tall object can be an optical illusion to the observer. This phenomenon is called mirage.

Q32. Define mirage

ANS. Mirage are optical illusions that appears when light rays bends, or refract through layers of air with varying temperature and density.

Mirage is especially common in hot deserts.

While moving in a bus or a car during a hot summer day, a distant patch of road, especially on a highway, appears to be wet.

2. DIAMOND

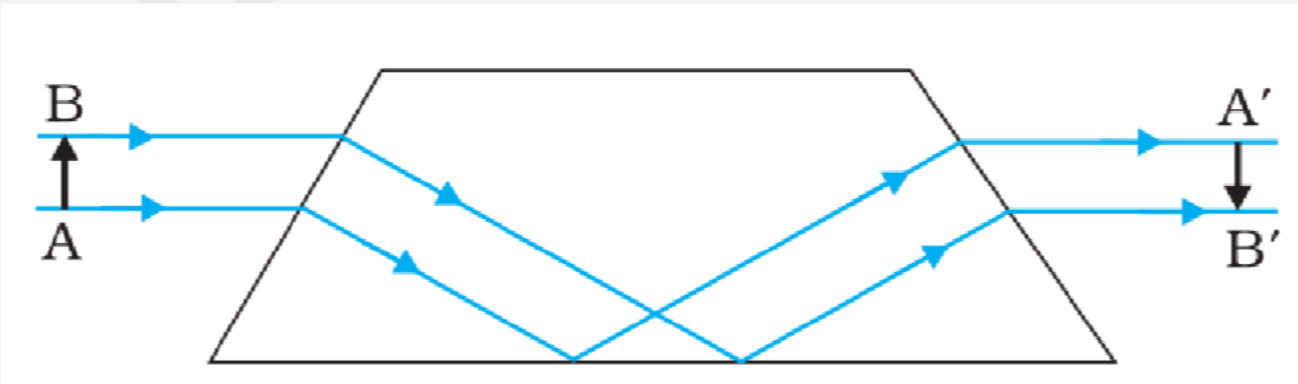
Diamond are known for their spectacular brilliance.

There brilliance is due to total internal reflection of light inside them.

- ➔ The critical angle for diamond-air interface (=24.4°) is very small, therefore once light enters a diamond it is very likely to undergo total internal reflection inside them.
- ➔ It is a technical skill of a diamond cutter which makes diamond to sparkle so brilliantly.
- ➔ by cutting the diamond suitably multi total internal reflection can be made to occur.

6. PRISM

- ➔ Prism designed to bend the light by 90° or 180° make use of total internal reflection. Such a prism is also used as an invert image without changing their size.

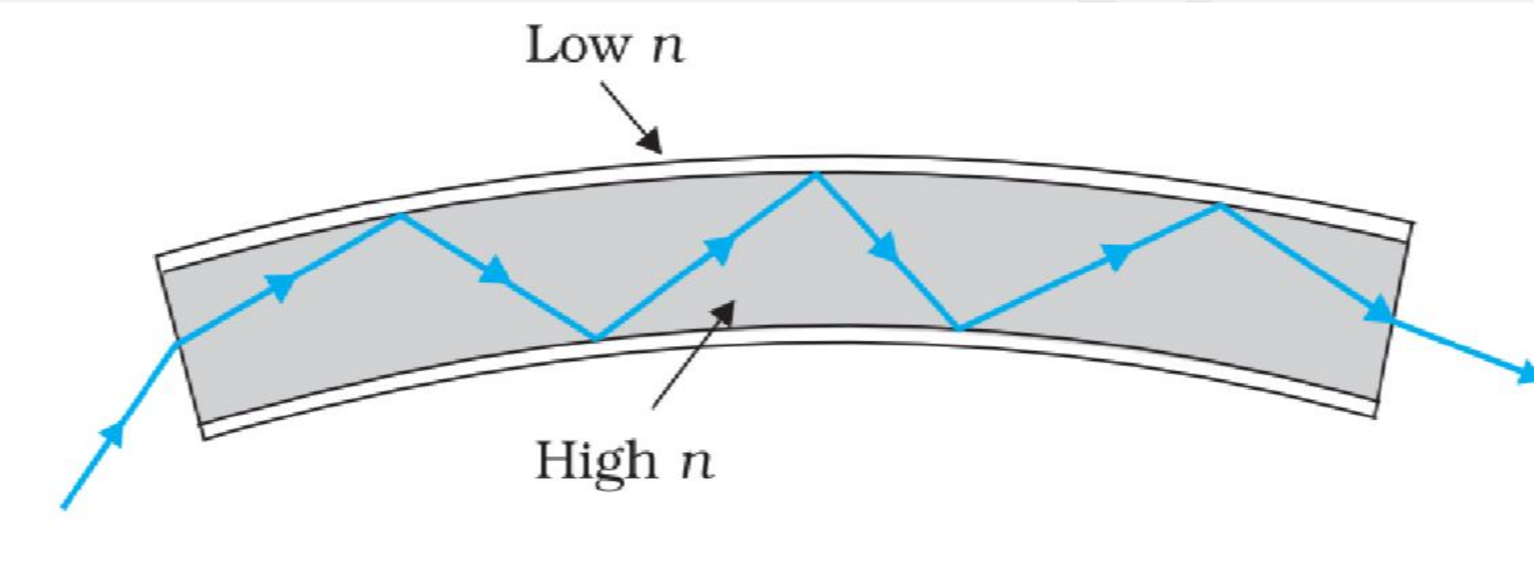


- ➔ The critical angle ic for the material of the prism must be less than 45°.
- ➔ This is true for crown glass and dense flint glass.

7. OPTICLE FIBRE

* Now a days optical fibres are extremely/exclusively used for transmitting audio and video signals through long distances.

- Optical fibres are extensively used for transmitting and receiving electrical signals which are converted to light by suitable transducers.
- Optical fibres are fabricated with high-quality composite glass/quartz fibres.
- Each fibre consists of a core and cladding.
- The refractive index of the material of the core is higher than that of the cladding.



- When a signal in the form of light is directed at one end of the fibre at a suitable angle, it undergoes repeated total internal reflections along the length of the fibre and finally comes out at the other end.
- Since light undergoes total internal reflection at each stage, there is no appreciable loss in the intensity of light signal.
- An optical fibre is fabricated such that light reflected from one side of its inner surface strikes the opposite inner surface at an angle greater than the critical angle.
- Even if the fibre is bent, light can easily travel along its length. Thus, an optical fibre can be used to act as a Pipe.
- Optical fibres can be used for the transmission of optical signals. (For example, they are used as light pipes to facilitate the visual examination of internal organs such as the oesophagus, stomach, and intestine.)
- Decorative lamps use fine plastic fibres, with their free ends forming a fountain-like structure. The other ends of the fibres are fixed over an electric lamp. When the lamp is switched on, light travels through each fibre and appears at the tip of its free end as a bright dot of light.
- The main requirement in fabricating optical fibres is that there should be very little absorption of light as it travels long distances through them.
-

- There is no appreciable loss in the intensity of the light signal. This has been achieved by the purification and special preparation of materials such as glass and quartz.
- In silica glass fibres, it is possible to transmit more than 95% of the light through a fibre of length 1 km.

REFRACTIVE INDEX AT SPHERICAL SURFACE & BY LENCES

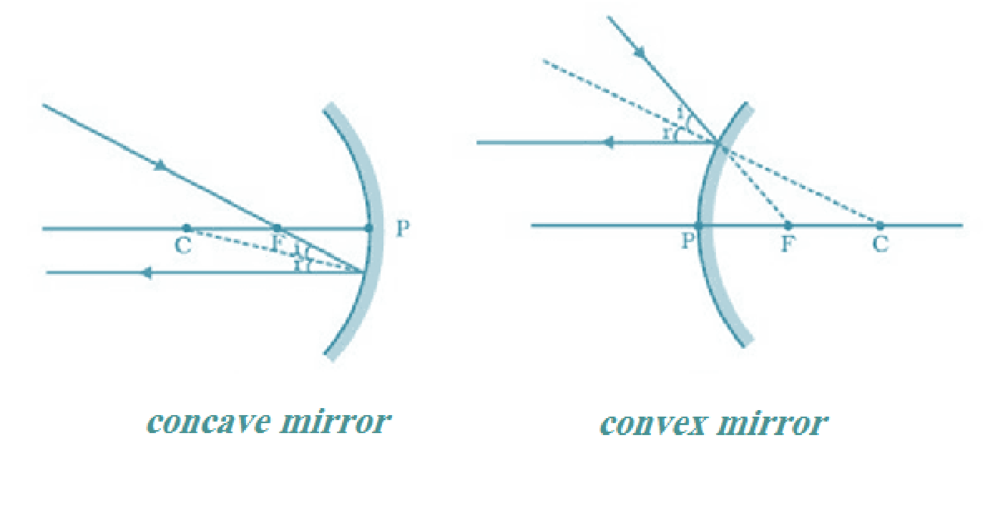
Refraction at a plane surface

a. *Glass slab*
b. *Prism*
c. *Diamond*

Refraction at a spherical surface

d. *Lens*
e. *Rainbow drop*
f. *Human eye*

- An infinitesimal part of a spherical surface can be regarded as planar and the same laws of reflection can be applied at every point on the surface.
- The normal at the point of incidence is perpendicular to the tangent plane to the spherical surface at that point and therefore passes through its centre of curvature.



- Refraction by a single spherical surface and follow it by thin lenses.

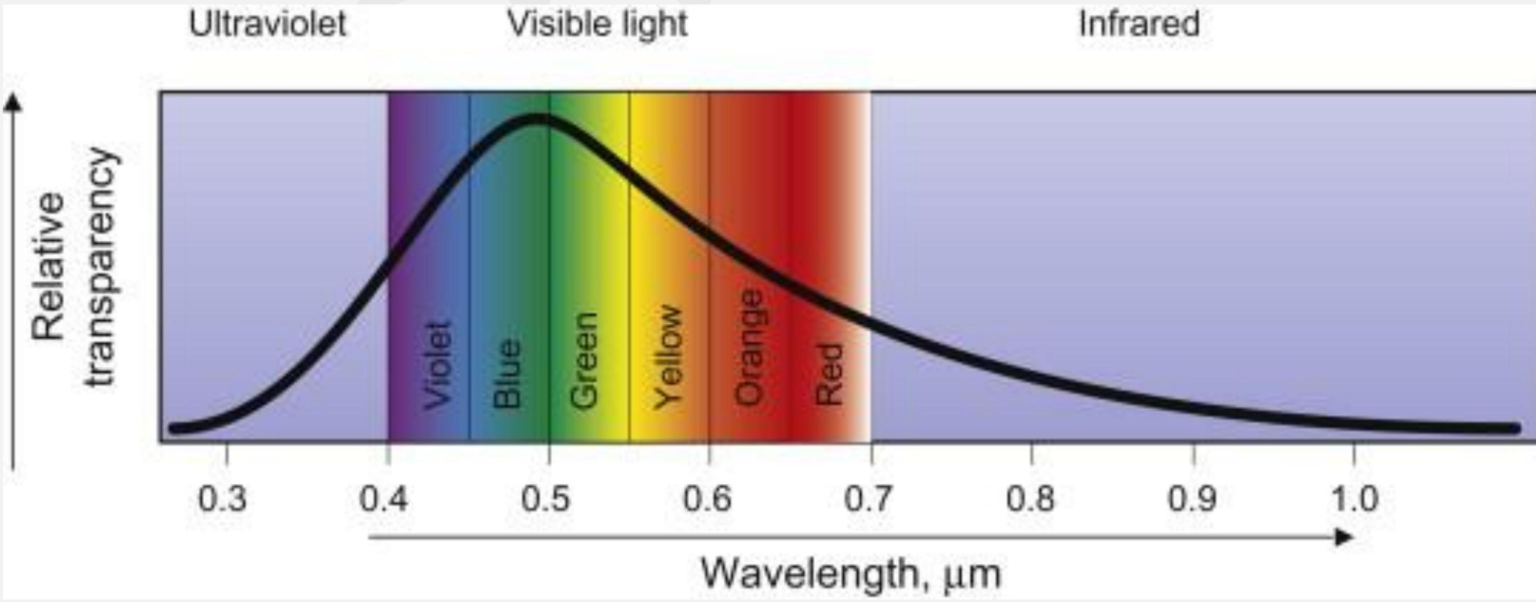
Q33. Define thin lens.

ANS. A thin lens. A transparent optical medium bounded by two surfaces, at least one of which should be spherical.

- Applying the formula for image formation by a single spherical surface successively at the two surfaces of a lens, we shall obtain the lens maker formula.

LIGHT SOURCES & PHOTOMETRY

- It is known that a body above absolute zero temp. emits E.M. radiation.
- The wavelength region in which the body emits the radiation, depends on its absolute temperature.
- Radiation emitted by a hot body, for example a tungsten filament lamp having temperature 2850 K are partly invisible and mostly in infrared region.
- As the temperature of the body increase radiation emitted by it is in the visible region.
-
- The sun with temperature of about 5500K emits radiation whose energy verses wavelength graph peaks approximately at 550 nm corresponding to green light and is almost in the middle of visible region.



- The energy verses wavelength distribution graph for a given body peaks at some wavelength which is inversely proportional to absolute temperature of that body.

Q.34 Define photometry.

ANS. **The measurement of light as perceived by human eye is called photometry.**

- Photometry is measured of a physiological phenomenon, being the stimulus of light as received by the human eye, transmitted by the optic nerves and analysed by the Brain.

The main physical quantities in photometry are-

g. *The luminous intensity of the source.*
h. *The luminous flux or flow of light from the source.*
i. *Illuminance of the surface.*

- The SI unit of luminous intensity is candela (cd).

Q35. Define One candela.

Ans. **The candela is the luminous intensity in a given direction of a source that emits monochromatic radiation having frequency 540 × 10^12 Hz and that has a radiant intensity in that direction of 1/683 watt per steradian.**

Q36. Define One lumen.

Ans. **If a light source emits one candela of luminous intensity into a solid angle of one steradian, the total luminous flux emitted into that solid angle is one lumen (lm).**

- A standard 100-watt incandescent light bulb emits approximately 1700 lumen.
- In photometry the only parameter which can be measured directly is illuminance.

Q37. Define illuminance. (lm/m^2 or lux)

Ans. **Illuminance is defined as luminous flux incident per unit area on a surface.**

i.e., Illuminance = Lumen / m^2 = Lux

- Most light meters measure illuminance.
- The illuminance **'E'** produced by a source of luminous intensity **'I'** is given by:
- **$E = I / r^2$**, where **'r'** is the normal distance of the surface from source.
- A quality named luminance (L) is used to characterise the brightness of emitting or reflecting flat surface.
- S.I. unit of luminance is Cd/m^2, sometimes called nit in industry.
- A good LCD computer monitor has a brightness of 250 nits.
- Aperture is also known as lateral size.

Refraction at a spherical surface: -

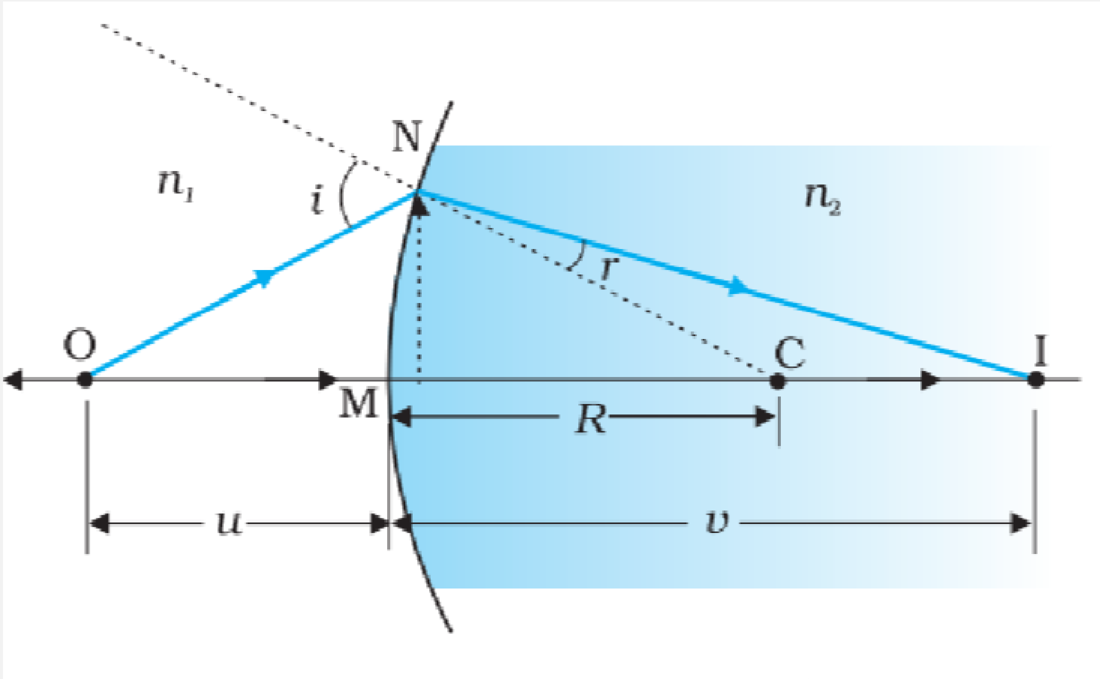


Fig: Refraction at a spherical surface separating two media.

→ For small angles,

$$\tan \angle NOM = \frac{MN}{OM}$$
$$\tan \angle NCM = \frac{MN}{CM}$$
$$\tan \angle NIM = \frac{MN}{MI}$$

Now, for ΔNOC, i is the exterior angle.

$$i = \angle NOM + \angle NCM$$
$$\tan i = \tan \angle NOM + \tan \angle NCM$$
$$\tan i = \frac{MN}{OM} + \frac{MN}{CM}$$

Similarly, in ΔNIC,

∠NCM is the exterior angle property.

$$\angle NCM = r + \angle NIM$$
$$r = \angle NCM - \angle NIM$$

now

$$r = \frac{MN}{CM} - \frac{MN}{MI} \ldots (ii)$$

Now, by Snell's law

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$$
$$n_1 \sin i = n_2 \sin r$$

For small angle,

$$\sin i = i$$

and

$$\sin r = r$$

Hence,

$$n_1 i = n_2 r$$
$$n_1\left(\frac{MN}{OM} + \frac{MN}{CM}\right) = n_2\left(\frac{MN}{CM} - \frac{MN}{MI}\right)$$
$$n_1 MN\left(\frac{1}{OM} + \frac{1}{CM}\right) = n_2 MN\left(\frac{1}{CM} - \frac{1}{MI}\right)$$
$$\frac{n_1}{OM} + \frac{n_1}{CM} = \frac{n_2}{CM} - \frac{n_2}{MI}$$

$$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{CM}$$

$$OM = -u, MI = +v, CM = +R$$

Substituting these values,

$$\frac{-n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$$

or,

$$\boxed{\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}}$$

→ This equation gives us a relation between object distance and image distance in terms of the refractive indices of the media and the radius of curvature of the spherical surface. This relation holds for any curved spherical surface.

Refraction by a lens

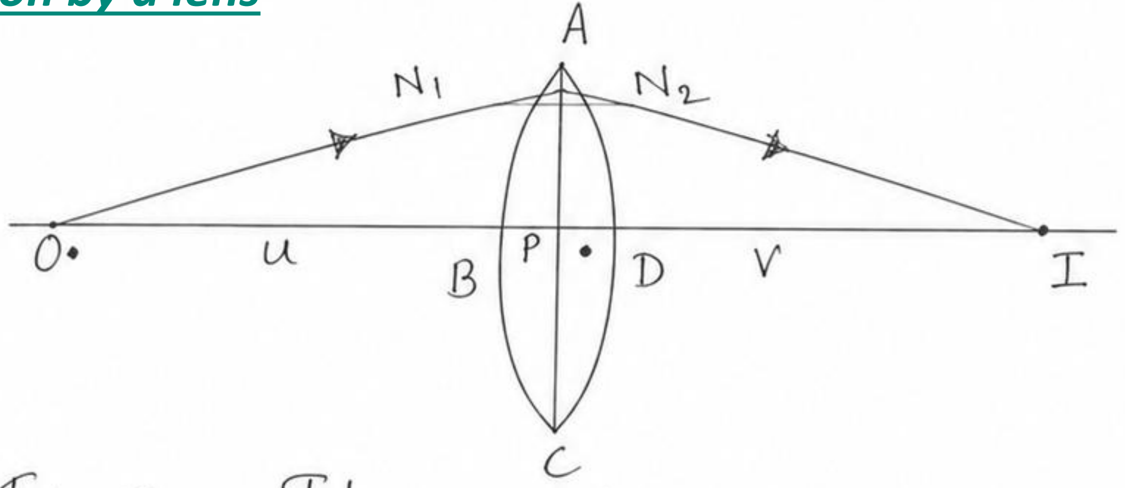


Fig ⓐ – The fasition of object, and the image, formed by a double convex lens

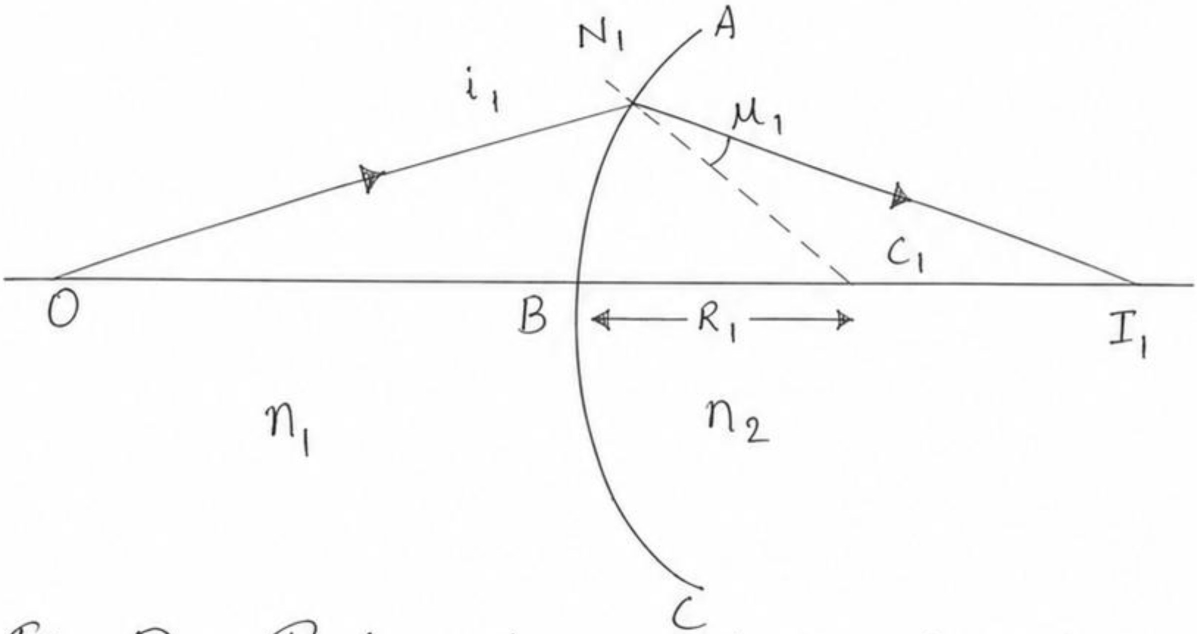
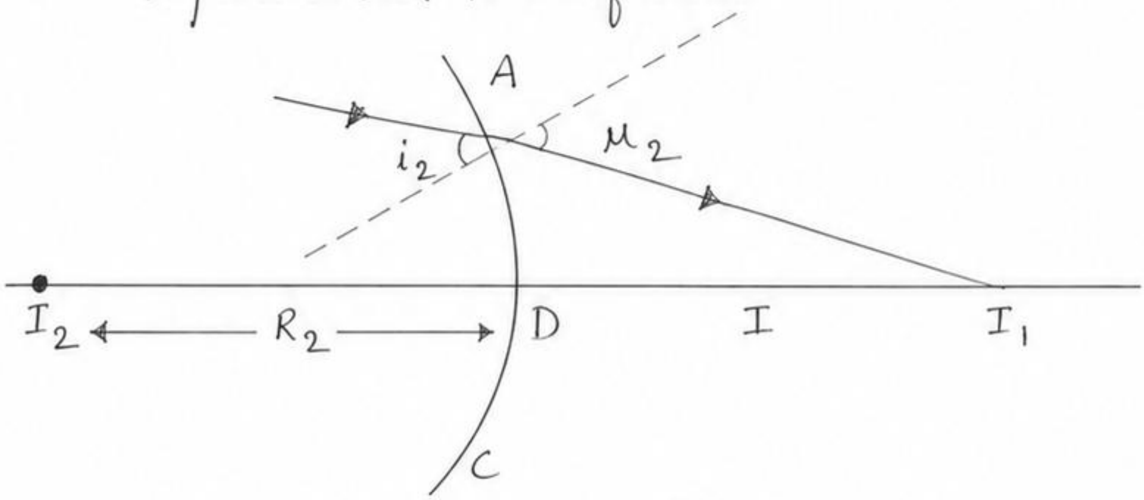


Fig Ⓑ – Refraction at the first spherical surface



Refraction at the second spherical surface

Fig (a) shows the geometry of image formation by a double convex lens.

The image formed by a double convex lens can be seen in terms of two steps:

(i)

The first refracting surface forms the image I_1 of the object O as shown in Fig. (b).

(ii)

The image I2 acts as a virtual object for the second surface that forms the image at I.

Now applying the equation,

$$\frac{n_1}{OB} + \frac{n_2}{BI_1} = \frac{n_2 - n_1}{BC_1} \ldots (i)$$

A similar procedure applied to the second interface ADC gives,

$$-\frac{n_2}{DI_1} + \frac{n_1}{DI} = \frac{n_2 - n_1}{DC_2} \ldots (ii)$$

For a thin lens,

$$BI_1 = DI_1$$

Adding (i) and (ii), we get—

$$\frac{n_1}{OB} + \frac{n_2}{BI_1} - \frac{n_2}{DI_1} + \frac{n_1}{DI} = \frac{n_2 - n_1}{BC_1} + \frac{n_2 - n_1}{DC_2}$$

Since,

$$BI_1 = DI_1$$

their terms cancel.

Therefore,

$$\frac{n_1}{OB} + \frac{n_1}{DI} = (n_2 - n_1)\left(\frac{1}{BC_1} + \frac{1}{DC_2}\right) \ldots (3)$$

Suppose the object is at infinity, i.e.,

$$OB \to \infty$$

and

$$DI = f$$

From (3),

$$\frac{n_1}{\infty} + \frac{n_1}{f} = (n_2 - n_1)\left(\frac{1}{BC_1} + \frac{1}{DC_2}\right)$$

Since,

$$\frac{n_1}{\infty} = 0$$

we get,

$$\frac{n_1}{f} = (n_2 - n_1)\left(\frac{1}{BC_1} + \frac{1}{DC_2}\right) \ldots (4)$$

→ The point where the image of an object placed at infinity is formed is called the focus (F) of the lens, and the distance f gives its focal length.

→ A lens has two foci, F and F_1, on either side of it.

BC1 = +R1

DC2 = -R2

From Equation (4),

$$\frac{n_1}{f} = (n_2 - n_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Dividing both sides by n_1,

$$\frac{1}{f} = \left(\frac{n_2 - n_1}{n_1}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Since,

$$\mu = \frac{n_2}{n_1}$$

therefore,

$$\boxed{\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}$$

This equation is known as the Lens Maker's Formula.

→ Lens maker's formula is useful to design lenses of desired focal length using surfaces of suitable radii of curvature.

Sign convention

For a convex lens

- $R_1 = +$
- $R_2 = -$
- $f = +$

For a concave lens

- $R_1 = -$
- $R_2 = +$
- $f = -$

Using Equation (3),

$$\frac{n_1}{OB} + \frac{n_1}{DI} = (n_2 - n_1)\left(\frac{1}{BC_1} + \frac{1}{DC_2}\right) \ldots (5)$$

From Equation (4),

$$\frac{n_1}{f} = (n_2 - n_1)\left(\frac{1}{BC_1} + \frac{1}{DC_2}\right) \ldots (6)$$

From (5) and (6),

$$\frac{n_1}{OB} + \frac{n_1}{DI} = \frac{n_1}{f}$$

Dividing by n_1,

$$\frac{1}{OB} + \frac{1}{DI} = \frac{1}{f}$$

Using the Cartesian sign convention,

$$OB = -u, DI = +v$$

Hence,

$$\frac{1}{f} = -\frac{1}{u} + \frac{1}{v}$$

or,

$$\boxed{\frac{1}{f} = \frac{1}{v} - \frac{1}{u}}$$

This is known as the Thin Lens Formula.

Q38. Define first focal point.

ANS. **The focus on the side of the (original) source of light is called the first focal point.**

Q39. Define second focal point.

ANS. **The focus on the other side of the source of light is called the second focal point.**

Q40. Define magnification (m) produced by a lens.

ANS. **Magnification (m) produced by a lens is defined as the ratio of the size (height) of the**
image to that of the object.

$$\boxed{m = \frac{h'}{h} = \frac{v}{u}}$$

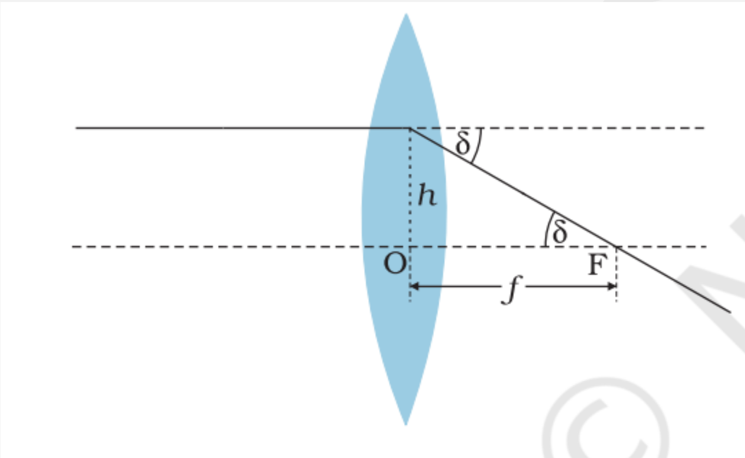
- If m is positive (+ve), the image formed is erect and virtual (as in a convex lens for certain object positions or a concave lens).
- If m is negative (-ve), the image formed is real and inverted.

Power of lens

- Power of a lens is a measure of the convergence or divergence which a lens introduces in the light falling on it.
- A lens of shorter focal length bends the incident light more.

Q41. Define power of a lens.

Ans. **The power of a lens is defined as the tangent of the angle by which it converges or diverges a beam of light falling at unit distance from the optical centre.**



$$\tan\delta = \frac{h}{f}, \text{if } h = 1$$

$$\therefore \tan\delta = \frac{1}{f}$$

For small value of δ,

$$\tan\delta = \frac{1}{f} \Rightarrow S = \frac{1}{f}$$

Thus,

$$P = \frac{1}{f}$$

S.I. unit for power of a lens is dioptre (D).

Q42. Define one dioptre.

Ans. The power of a lens of focal length 1 metre is one dioptre.

- Power of lens is positive for a converging lens (convex lens).
- Power of lens is negative for a diverging lens (concave lens).

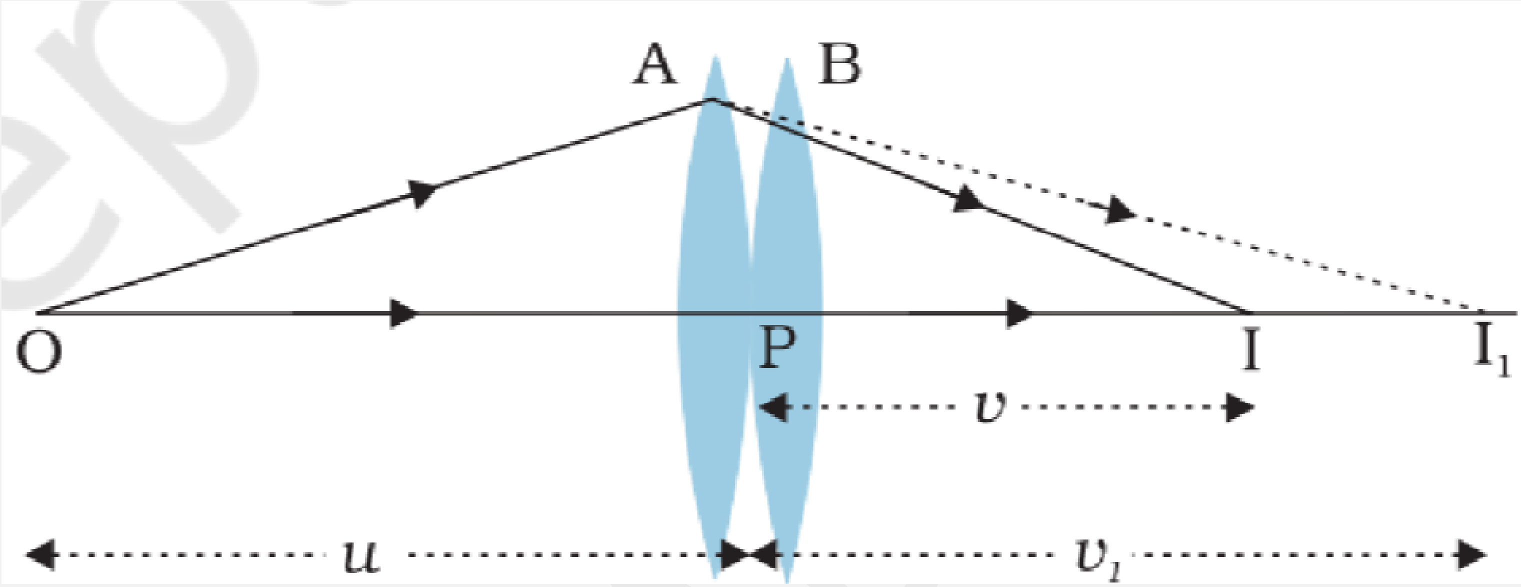


Fig: Image formation by a combination of thin lenses in contact.

- The first lens produces an image at I_1. Since the image I_1 is real, it serves as a virtual object for the second lens B, producing the final image at I.
- The formation of the image by the first lens is assumed only to facilitate the determination of the position of the first image.
- In fact, the direction of the rays emerging from the first lens is modified according to the angle at which they strike the second lens.
- Since the lenses are thin, we assume the optical centres of the lenses to be coincident.
- Let the central point be denoted by P.

For the image formed by the first lens A, we get

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} (i)$$

For the image formed by the second lens B, we get

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} (ii)$$

Adding (i) and (ii), we get

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f_1}+\frac{1}{f_2}$$

Hence,

$$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$$

If the two Lens-system is regarded as equivalent to a signal lens of focal length 'f', we get: -

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

The derivation is valid for any number of thin lenses in contact.

If several thin lenses of focal lengths $(f_1, f_2, f_3, \ldots)$ are in contact, the effective focal length of their combination is given by

$$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}+\cdots$$

In terms of power,

$$P=P_1+P_2+P_3+\cdots(3)$$

where $P$is the net power of the lens combination.

In equation (3), the algebraic sum of individual powers is taken. Therefore, some terms on the right side may be **positive** (for convex lenses) and some **negative** (for concave lenses).

- Combination of lenses helps to obtain diverging or converging lenses of desired magnification. It also enhances the sharpness of the image.
- The image formed by the first lens becomes the object for the second lens.

  Since

$$P=\frac{1}{f},$$

it implies that the total magnification $m$of the combination is the product of the magnifications $(m_1, m_2, m_3, \ldots)$of the individual lenses.

$$m=m_1\times m_2\times m_3\times\cdots$$

- Such a system of combination of lenses is commonly used in designing lenses for **cameras, microscopes, telescopes, and other optical instruments**.

Refraction through A prism

Q43. Define angle of deviation δ

ANS. The angle b/w the emergent ray and the direction of the incident ray is called the angle of deviation, δ.

$\angle A + \angle ONR = 180° — (i)$

From the triangle, ONR,

$r_1 + r_2 + \angle ONR = 180° — (ii)$

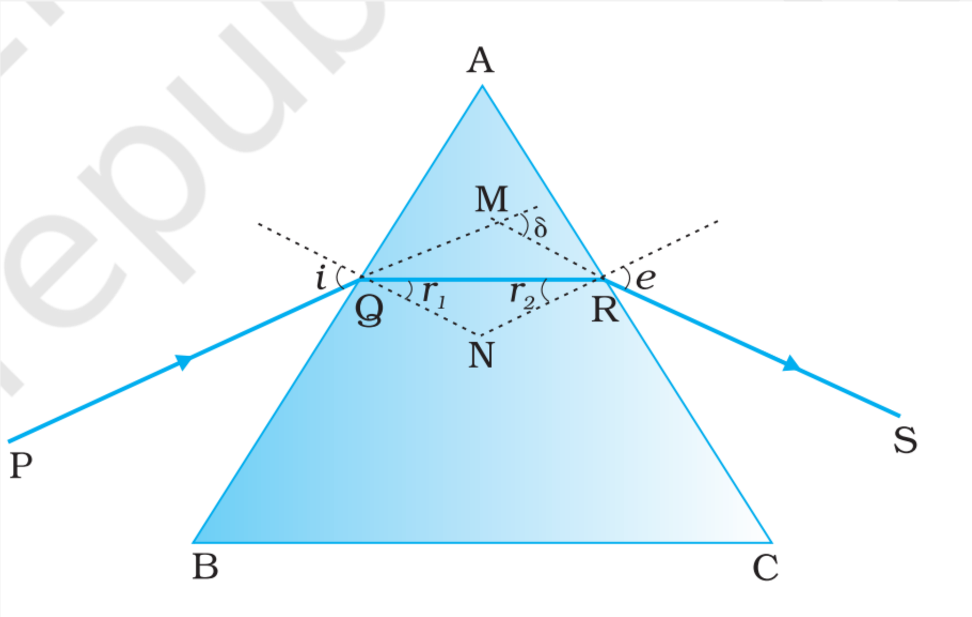


FIGURE 9.21 A ray of light passing through a triangular glass prism.

From (i) and (ii), we get: -

$\angle A + \angle ONR = r_1 + r_2 + \angle ONR$

$\angle A = r_1 + r_2$

The total deviation δ is the sum of deviations at the two faces,

$\delta = (i - r_1) + (e - r_2)$

i.e., $\delta = i + e - (r_1 + r_2)$

$\delta = i + e - A$

↳ *The angle of deviation depends on the angle of incident.*

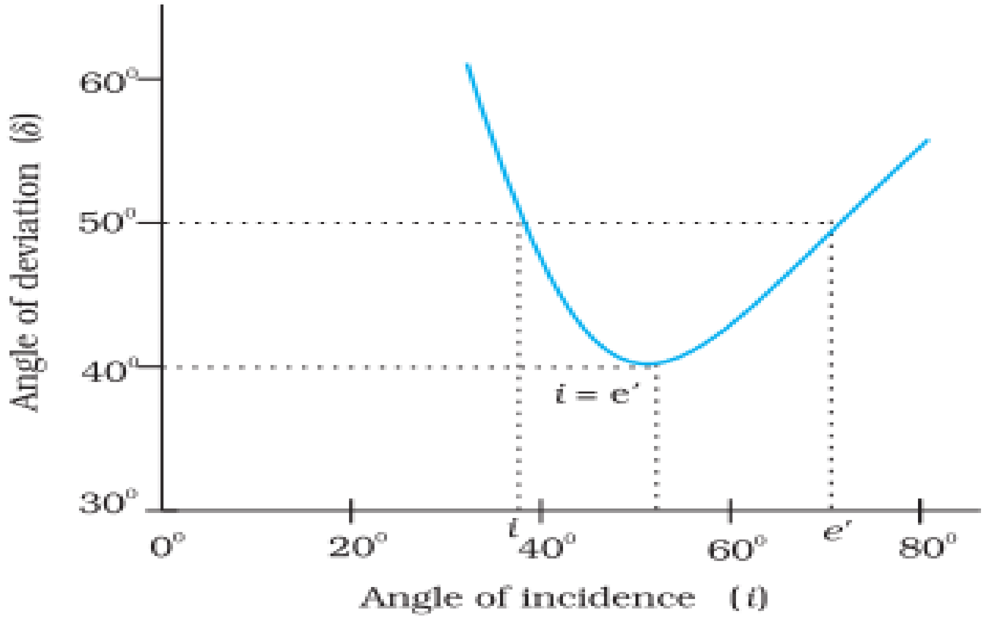


FIGURE 9.22 Plot of angle of deviation (δ) versus angle of incidence (i) for a triangular prism.

- ➔ Any given of δ, except for i = 0 corresponds to two values of i and have of e:
- ➔ This in fact, is expected from the symmetry of i' and e' in eq

δ = i + e - A remains same if i' and e' are inter changed.

At the minimum deviation Dm, the refracted ray inside the prism becomes parallel to its base.

$\delta = Dm$; $i = e$ which implies
or $i = r_2$

$r_1 + r_2 = A$
$r + r = A$
$2r = A$

$r = A/2$

$Dm = \delta = i + e - A$

$Dm + A = i + e$
$Dm + A = i + i$
$Dm + A = 2i$

$$i = Dm + \frac{A}{2}$$

The refractive index of the prism

$$m_{21} = m_2/m_1 = \frac{sini}{sinr}$$

$$m = \frac{sin\left[\frac{A + Dm}{2}\right]}{\frac{sinA}{2}}$$

For a small angle prism, i.e; A then prism Dm is also very small, we get -

$$m_{21} = sin\,[(A + Dm)/2] \,/\, sin\left[\frac{A}{2}\right]$$

$$= \frac{\frac{[A+Dm]}{2}}{\frac{A}{2}} \quad = A + \frac{Dm}{A}$$

$$m_{21} = A + Dm\,/\,A$$

$$m_{21}\,A = A + Dm$$

$$(m_{21} - 1)\,A = Dm$$

$$Dm = A\,(\mu - 1)$$

It implies that, thin prism does not deviate light much.

Dispersion by a prism

- ➔ When a narrow beam of sunlight, usually called white light, is incident on a glass prism, the emergent light seems to be consisting of several colours.
- ➔ There is actually a continuous variation of colour, but broadly, the different component colours that appears in sequences are violet, Indigo, Blue, Green, Yellow, Orange and Red (given by the acronym VIBGYOR).
- ➔ The red light bends the least and the violet light bends the most.

Q44. Define Dispersion:

ANS. The phenomenon of splitting of light in to its component/Component colours is known as dispersion.

Q45. Define spectrum of light

ANS. The pattern of colour components of light is called the spectrum.

- ➔ The inverted prism recombines them to give white light.
- ➔ White light itself consists of light of different colours which are separated by the prism.
- ➔ A ray of light as defined mathematically doesn't exist.
- ➔ An actual ray is really a beam of many light rays.
- ➔ Each ray splits into component colours when it enters the glass prism.
- ➔ Colour is associated with the wavelength of light.
- ➔ In the visible spectrum, red light is at the long wavelength end (~700 nm) while the violet light is at short wavelength end (~400 nm)

- ➔ Dispersion takes place because the refractive index of medium for different wavelength is different.
- ➔ For e.g.; the bending of red component of white light is least, while it is most for the violet.
- ➔ Red light travels faster than violet light in glass prism.

- Thick lens could be assumed as made of many prisms; therefore, thick lenses show chromatic aberration due to dispersion of light.
- The variation of refractive index with wavelength may be more pronounced in some media than the other.
- In vacuum speed of light independent of wavelength.
- vacuum (or air approximately) is a non-dispersive medium, in which, all colours' travels with the same speed. This also follow from the fact that sunlight reaches us in the form of white light and not as its component.
- Glass is a dispersive medium.

Some Natural Phenomenon Due to Sunlight Are: -

i. The blue colour of the sky
ii. White clouds
iii. The Red hue at sunrise and sunset
iv. The Rainbow
v. The brilliant colours of some pearls
vi. Shells
vii. Wings of Birds.

The Rainbow

- ➔ The rainbow is an example of the dispersion of sunlight by the water drops in the atmosphere.
- ➔ Rainbow is a phenomenon due to combined effect of dispersion, refraction and reflection of sunlight by spherical water droplets by rain.
- ➔ The conditions for observing a rainbow are that the Sun should be shining in one part of the sky, say near the western horizon, while it is raining in the opposite part of the sky, say near the eastern horizon.
- ➔ An observer can, therefore, see a rainbow only when his back is towards the Sun.
- ➔ The formation of rainbow as sunlight is first refracted as it enters a spherical raindrop, which causes the different wavelength (colours) of white light to spread.
- ➔ longer wavelength of light (red) is bent the least while short wavelength (violet) is bent the most.
- ➔ These component rays strike the inner surface of the water drops and get internally reflected, if the angles b/w the refracted ray and normal to the drop surface is greater than the critical angle (48° in this case)

- ➔ The reflected light is refracted again as it comes out of the drop.
- ➔ The violet light emerges at an angle of 40° and red light emerges 42° related to the incoming sunlight.

Types of Rainbows

1. ***Primary Rainbow***
2. ***Secondary Rainbow***

Formation of Primary Rainbow

- ➔ Red light from drop1 and violet light from drop2 reach the observer's eye.
- ➔ The violet ray light from drop1 and red light from drop2 are directed at level or below the observer.
- ➔ Thus, the observer sees a rainbow with red colour on the top and violet on the bottom.
- ➔ The primary rainbow is a result of three-step process, that is Refraction, Reflection and Refraction.

Formation of secondary Rainbow

- ➔ Violet ray from drop1 and red light from drop2 reach the observer eye.
- ➔ Red light from drop1 and violet light from drop2 are directed at level or below the observer.

- Two internal reflections inside the rain drop.

- ➔ Secondary rainbow is a four-step process that are refraction, Reflection, Reflection and refraction
- ➔ Secondary rainbow is fainter than the primary rainbow.
- ➔ In secondary rainbow, violet colour on the top and red colour on the bottom.

Scattering of light

- ➔ As sunlight travels through the earth's atmosphere, it gets scattered (changes its direction) by the atmospheric particles.
- ➔ Light of shorter wavelength scatter much more than the light of longer wavelength.

Q46. Define Ray light scattering

- **ANS. The amount of scattering is inversely proportional to the fourth power of the wavelength** i.e.,

Scattering $\propto 1/\lambda^4$

$\theta \propto 1/\lambda^4$

- Hence, the bluish colour predominates in a clear sky, since blue has shorter wavelength than red and it is scattered much more strongly.

➔ Even violet gets scattered even more than blue, having a shorter wavelength. But, since our eyes are more sensitive to blue than violet, we see the blue.

➔ Large particles like dust and water droplets present in the atmosphere behave differently.

➔ The important quantity is the relative size of the wavelength of light, and the typical size, of the scatterer.

➔ For $a \ll \lambda$, one has Rayleigh scattering, which is proportional to $\frac{1}{\lambda^4}$

➔ For $a \gg \lambda$, i.e., large scattering objects (for example, raindrops), this is not true; all wavelengths are scattered nearly equally.

➔ Thus, clouds which have droplets of water with $a \gg \lambda$are generally white.

- At sunset or sunrise, the sun's rays have to pass through a longer distance in the atmosphere.
- Most of the blue and other short wavelengths are removed by scattering. The least scattered light reaches our eye.
- Therefore, the sun looks reddish.
- This explains the reddish appearance of the sun and full moon near the horizon.

Optical Instruments

➔ A number of optical devices and instruments have been designed utilising the reflecting and refracting properties of mirrors, lenses and prisms.

THE EYE

Q47. Define Cornea.

ANS. **Light enters the eye through a curved front surface called the cornea.**

➔ Light passes through the pupil, which is the central hole in the iris.

The size of the pupil can change under the control of muscles (**ciliary muscles**).

The light is further focused by the **eye lens** on the **retina**.

Q48. Define Retina.

ANS. **The retina is a film of nerve fibres covering the curved back surface of the eye.**

- The retina contains rods and cones, which sense light intensity and colour respectively, and transmit electrical signals via the optic nerve to the brain, which finally processes this information.
- The shape (curvature) and therefore the focal length of the lens can be modified somewhat by the **ciliary muscles**. For example, when the muscles are relaxed, the focal length is about **2.5 cm**, and objects at infinity are in sharp focus on the retina.
- **Image lens distance ≅ 2.5 cm**

Q49. Define Accommodation.

ANS. When an object is brought closer to the eye, in order to maintain the same image-lens distance (≈ 2.5 cm), the focal length of the eye lens becomes shorter by the action of the ciliary muscles. This property of the eye is called accommodation.

- If the object is too close to the eye, the lens cannot curve enough to focus the image on the retina, and the image is blurred.

Q50. Define least distance of distinct vision or near point.

ANS. The closest distance for which the eye lens focuses light on the retina is called the least distance of distinct vision or the near point.

- The standard value for normal vision is taken as 25 cm for the near point. It is denoted by D.
- Near point distance increases with age because of the decreasing effectiveness of the ciliary muscles and the loss of flexibility of the lens.
-
- The near point may be as close as about 8 cm in a child of about 10 years of age.
-
- The near point increases to as much as 200 cm at 60 years of age.

Q51. Define Presbyopia.

ANS. If an elderly person tries to read a book at about 25 cm from the eye, the image appears blurred. This condition is called presbyopia.

- Presbyopia is corrected by using a converging (convex) lens for reading.

Common optical defects of the eye

i) Myopia (Near-sightedness)
ii) Hypermetropia (Far-sightedness)
iii) Astigmatism

Q52. Define Myopia.

ANS. The light from a distant object arriving at the eye lens may get converged at a point in front of the retina. This type of defect is called near-sightedness (myopia).

- This means that the eye is producing too much convergence in the incident beam.
- To compensate this, we interpose a concave lens between the eye and the object, with the diverging effect desired to get the image focused on the retina.

Q53. Define farsightedness or hypermetropia.

ANS. If the eye lens focuses the incoming light at a point behind the retina, a converging (convex) lens is needed to compensate for the defect in vision. This defect is called farsightedness or hypermetropia.

Q53. Define Astigmatism.

ANS. Astigmatism occurs when the cornea is not spherical in shape. For example, the cornea may have a larger curvature in the vertical plane than in the horizontal plane, or it may have a larger curvature in the horizontal plane than in the vertical plane.

- If a person having astigmatism looks at a wire mesh or a grid of lines, focusing on either the vertical or the horizontal lines, one set of lines may not appear as sharp as the other.
- Astigmatism results in lines in one direction being well focused, while those in the perpendicular direction may appear distorted.
- Astigmatism can be corrected by using a cylindrical lens of the desired radius of curvature with an appropriately directed axis.
- Astigmatism can occur along with myopia or hypermetropia.

OPTICAL INSTRUMENT

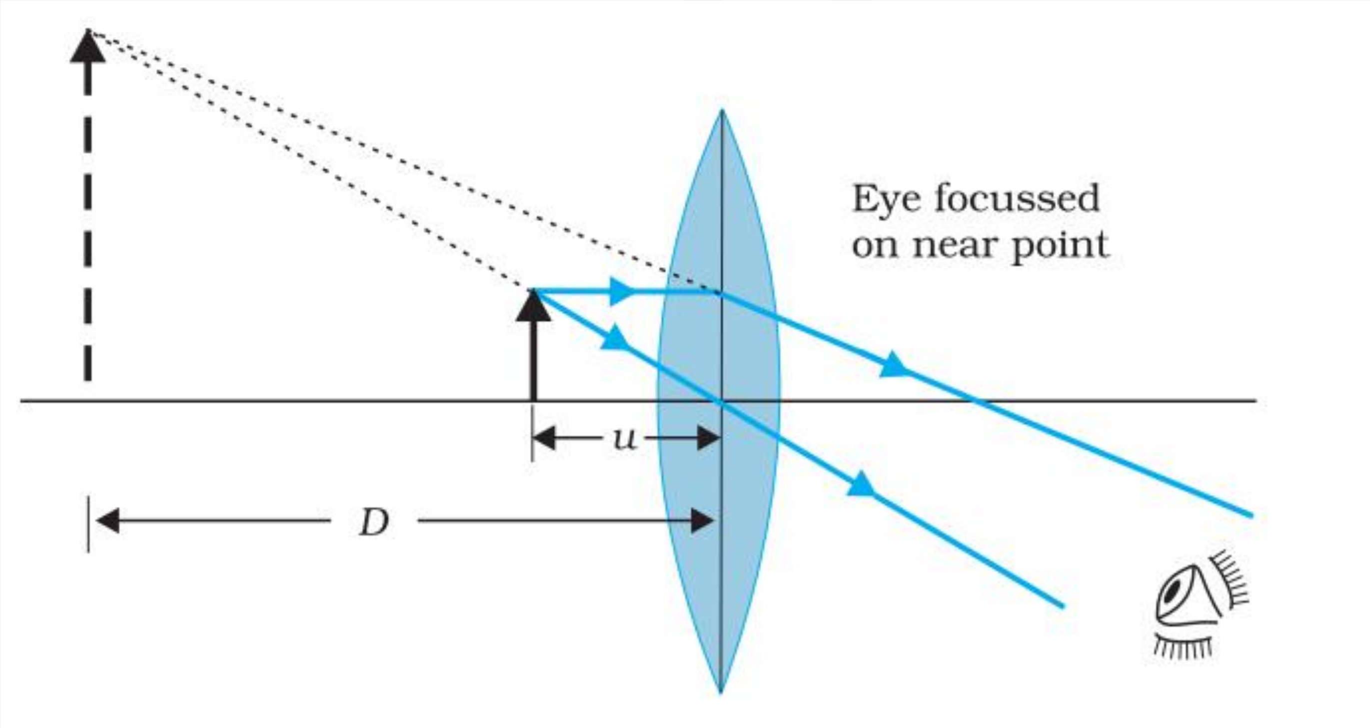
➔ A number of optical devices and instruments have been designed utilising reflecting and refracting properties of mirrors, lenses and prisms.

Examples of optical devices and instruments are: -

3. *a) Periscope*
   b) Kaleidoscope
   c) Binoculars
   d) Telescopes
   e) Microscopes.

#The Microscopes

→ A simple magnifier or microscope is simply converging lens of small focal length.



➔ In order to use such a lens as a microscope, the lens is held near the object, within its focal length, and the eye is positioned close to the lens on the other side.

➔ The idea is to get an erect magnified and virtual image of the object at a distance so that it can be viewed comfortably, i.e. at 25 cm or more.

CASE I

If the object is at a distance f, the image is at infinity.

Case II

If the object is at a distance slightly less than the focal length of the lens, the image is virtual and closer than infinity.

CASE III

The most comfortable distance for viewing the image is when it is at the near point (distance D = 25 cm), it causes some strain on the eye.
Therefore, the image formed at infinity is often considered to be most suitable for viewing by the relaxed eye.

The linear magnification for the image formed at the same point D, by a simple microscope can be obtained by using the relation: -

$$m = v/u = v(1/v - 1/f) = \left(1 - \frac{v}{f}\right)$$

$$V' = \text{-ve},\ V = D = 25\text{cm}$$

$$m = \left(1 + \frac{D}{f}\right)$$

$D = 25\text{cm}.$

$m = 6$

$$m = 1 + \frac{D}{f}$$

$$6 = 1 + \frac{25}{f}$$

$$6 - 1 = \frac{25}{f}$$

$$5 = \frac{25}{f}$$

$$f = \frac{25}{5}$$

$$f = 5cm$$

$m = \frac{h'}{h}$, *where*

h = size of the object

h' = size of the image

MAGNIFICATION

"The angular magnification is the ratio of the angle subtended by the image at the eye to the angle subtended by the object at the eye, when the object is viewed directly at the near point D.

- What a single lens simpler magnifier, achieves is that it allows an object to be brought closer to the eye than D.

Finding the magnification when the image is at infinity.

- In this case, we will have to obtain the angular magnification.
- The angle subtended is then given by

$Tan\ \theta 0 = \frac{h}{D} = \theta 0$

$$m = \frac{h'}{h} = \frac{v}{u}$$

The angle subtended by the image

$$\tan \theta i = \frac{h'}{-v}$$

$$tan\theta i = \frac{h'}{-v} \times \frac{v}{u} = \frac{h}{-v} = \theta$$

The angle subtended by the object, when it Is at u = -f

$$\theta i = \frac{h}{f}$$

The angular magnification is therefor,

$$m = \left(\frac{\theta i}{\theta°}\right) = \frac{D}{f}$$

- This is one less than the magnification when the final image is formed at the near point, but the viewing is more comfortable and the difference in magnification is usually small.
- Optical instruments (microscope and telescope) we shall assumes the image to be at Infinity.
- A simple microscope as a limited by maximum Magnification (≤ 9) for realistic focal length.

Q. Define Compound Microscope.

ANS. For much larger magnifications, two lenses are used, one compounding the effect of the other. Such an instrument is known as a compound microscope.

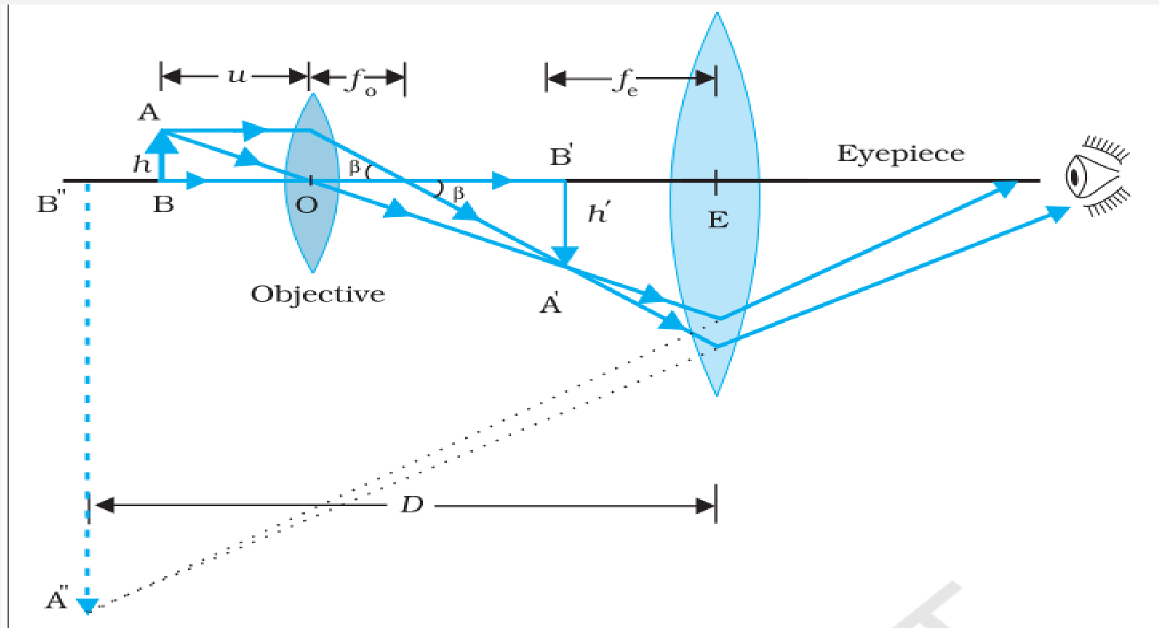


FIGURE 9.24 Ray diagram for the formation of image by a compound microscope.

Objective lens

The lens nearest the object is called the objective. It forms a real, inverted and magnified image of the object.

Eye piece

This is the second lens. It serves as a magnifier and produces the final enlarged and virtual image.

- ➔ The first inverted image is thus near (at the within) the focal plane of the eye piece at a distance Appropriate for final image formation add D Infinity or a Closer for image formation at the near point.
- ➔ Clearly the final image is inverted with respect to the original object.

Q. Obtain an expression of magnification due to a compound microscope.

ANS. The ray diagram shows that the (linear) magnification due to the objective, namely h'/h equals

$$m_0 = h'/h = \frac{L}{f_0}$$

$tan\ \beta = \left(\frac{h'}{f_0}\right) \approx \left(\frac{h'}{L}\right) where,$

$h' = Size\ of\ the\ first\ image$

$h = Object\ size$

$f_0 = focal\ length\ of\ the\ objective$

- The first image is formed near the focal point of the eyepiece.
- The distance L, i.e. the distance b/w the second focal point of the objective and the first focal point of the eyepiece (for a simplified f_0) is called the tube length of the compound microscope.
- The first inverted image is near the focal point of the eyepiece.

For the simple microscope to obtain the (angular) magnification m_e due to it

$m_e = \left(1 + \frac{D}{f_e}\right)$, when the final image is formed at the near point, is

$$m_e = \left(1 + \frac{D}{f_e}\right)$$

- When the final image is formed at infinity the angular magnification due to the eyepiece.

$$m_e = \frac{D}{f_e}$$

- The total magnification, when the image is formed at infinity, is

$$m = m_0 m_e = \left(\frac{L}{f_0}\right)\left(\frac{D}{f_e}\right)$$

- To achieve a large magnification of a small object, the objective and eyepiece should have small focal lengths.
- In practice, it is difficult to make the focal length much smaller than 1 cm. Also large lenses are required to make L large.
- For eg, with an objective with f_0 = 1.0cm, and an eyepiece with focal length f_e = 2.0cm and a tube length of 20cm, the magnification is

$$m = m_0 m_e = \left(\frac{L}{f_0}\right)\left(\frac{D}{f_e}\right)$$

$$= \frac{20}{1} \times \frac{25}{2}$$

$$m = 250$$

- Various other factors such as illumination of the object contribute to the quality and visibility of the image.
- In modern microscopes, multiple multi-component lenses are used for both the objective and the eyepiece to improve image quality and minimise various optical aberrations (defects) in lenses.

Telescope

- The telescope is used to provide angular magnification of distant objects.
- Telescope also has an objective and an eyepiece.
- The objective has a large focal length and a much larger aperture than the eyepiece.
- Light from a distant object enters the objective, and a real image is formed in the tube at its second focal point.
- The eyepiece magnifies this image producing a final inverted image.
- The magnifying power m is the ratio of the angle β subtended at the eye by the final image to the angle α at which the object subtended at the lens or the eye. Hence,

$$m \approx \beta/\alpha \approx h/f_e = \frac{f_o}{f_e}$$

- In this case, the length of the telescope tube is $fo + fe$.
- Terrestrial telescopes have an additional pair of lenses (erecting lenses) to make the final image erect.
- Refracting telescopes can be used for terrestrial and astronomical observations.
- For example, consider a telescope whose objective has a focal length of 100 cm and the eyepiece has a focal length of 1 cm. The magnifying power of this telescope is

$$m = \frac{fo}{fe} = \frac{100cm}{1cm} = 100$$

- Let us consider a pair of stars of what separation, 1 arc minute of arc. The stars appear as though they are separated by an angle of $100 \times 1' = 100' = 1.67°$
- The main considerations with an astronomical telescope are its light-gathering power and its resolving power.
- The former depends on the area of the objective.
- With larger diameters, fainter objects can be observed.
- The resolving power or the ability to observe two objects distinctly, which are very nearly in the same direction, also depends on the diameter of the objective.
- The desirable aim in optical telescopes is to make them with the objective of large diameters.
- The largest lens objective in use has a diameter of 40 inch or 1.02 m. It is at the Yerkes Observatory in Wisconsin, USA.

****** Such big lenses tend to be very heavy and therefore difficult to make and support by their edges.***

- Modern telescopes use a curved mirror rather than a lens for the objective.

Reflecting Telescope: -

"Telescopes with mirror objectives are called reflecting telescopes.

#Reflecting telescopes have several advantages: -

- There is no chromatic aberration in a mirror.
- If a parabolic reflecting surface is chosen, spherical aberration is also removed.

➔ Mechanical support is much less of a problem since a mirror weighs much less than a lens of equivalent optical quality, and can be supported over its entire back surface, not just over its rim.

➔ One obvious problem with a reflecting telescope is that the objective mirror forms the image inside the telescope tube.

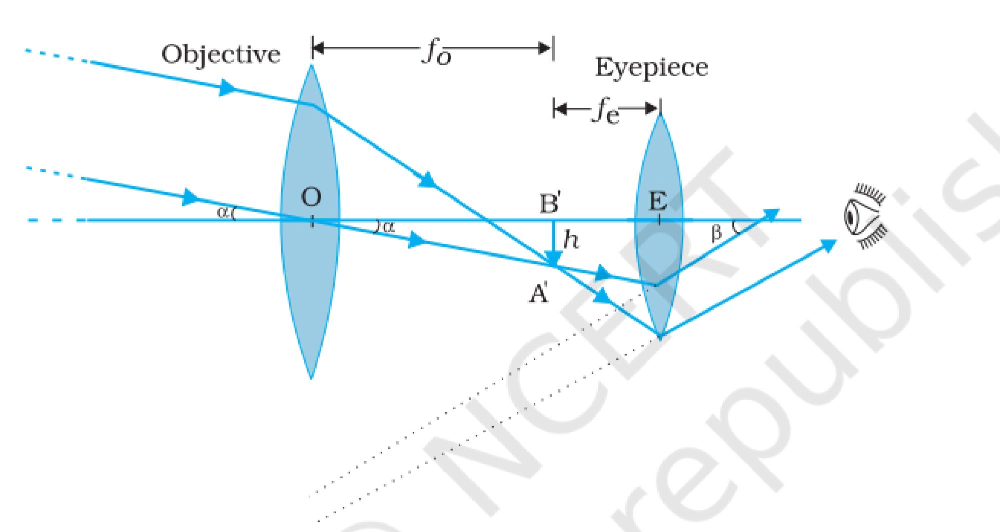


FIGURE 9.25 A refracting telescope.

Q. Define Cassegrainian telescope.

ANS. A Cassegrain telescope is a reflecting telescope, not a refracting telescope In this telescope, the objective lens is replaced by a large-aperture concave parabolic mirror of large aperture which is free from chromatic and spherical aberrations.

➔ The image formed is much brighter and the reflecting type telescope has much higher resolving power compared to the refracting type telescope. Such a telescope is known as Cassegrainian telescope.

➔ A Cassegrain telescope is a type of reflecting telescope.

➔ Cassegrain telescope has the advantages of a large focal length in a short telescope.

➔ It is a 2.34 m diameter reflecting telescope.

➔

➔ The largest reflecting telescopes in the world are the pair of Keck telescopes in Hawaii, USA, with a reflector of 10 metres in diameter.